

Q #1

Multiple Choice Type

Award: 1

Penalty: 0.33

Probability

Let X be a continuous random variable with probability density function:

$$f_X(x) = 3x^2 \quad \text{for } 0 \leq x \leq 1.$$

Define a new random variable $Y = \sqrt{X}$.

What is the p.d.f. of Y ?

A. $g(y) = 3y^2, \quad 0 \leq y \leq 1$

B. $g(y) = 6y^5, \quad 0 \leq y \leq 1$

C. $g(y) = 6y^4, \quad 0 \leq y \leq 1$

D. $g(y) = 9y^4, \quad 0 \leq y \leq 1$

Your Answer:

Correct Answer: B

Not Attempted

Time taken: 00min 23sec

Discuss

Q #2

Multiple Choice Type

Award: 1

Penalty: 0.33

Probability

Let U be a continuous random variable uniformly distributed over the interval $[-1, 2]$. What is the value of the expected value $E[U(U - 1)]$?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. 0

D. $\frac{2}{3}$

Your Answer:

Correct Answer: B

Not Attempted

Time taken: 00min 03sec

Discuss

Q #4

Multiple Choice Type

Award: 1

Penalty: 0.33

Probability

Let X_1 and X_2 be i.i.d. Poisson random variables with mean 1. Then $P(\max(X_1, X_2) > 1) =$

A. $1 - e^{-2}$

B. $1 - 2e^{-2}$

C. $1 - 3e^{-2}$

D. $1 - 4e^{-2}$

Your Answer:

Correct Answer: D

Not Attempted

Time taken: 00min 02sec

Discuss

Q #5

Multiple Select Type

Award: 1

Penalty: 0

Probability

Let X be a random variable with cumulative distribution function:

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ x^2 + \frac{1}{4}, & \text{if } 0 \leq x < \frac{1}{2} \\ x + \frac{1}{8}, & \text{if } \frac{1}{2} \leq x < \frac{3}{4} \\ \frac{x+1}{2}, & \text{if } \frac{3}{4} \leq x < 1 \\ 1, & \text{if } x \geq 1 \end{cases}$$

Which of the following is/are correct?

A. $P\left(0 \leq X < \frac{1}{4}\right) = \frac{5}{16}$

B. $P\left(X \geq \frac{3}{4}\right) = \frac{1}{8}$

C. $P\left(X = \frac{1}{2}\right) = \frac{1}{8}$

D. $P\left(0 \leq X < \frac{1}{4}\right) = \frac{1}{16}$

Your Answer:

Correct Answer: A;B;C

Not Attempted

Time taken: 00min 06sec

Discuss

Q #6

Multiple Choice Type

Award: 2

Penalty: 0.67

Probability

Let $U_1, U_2, \dots, U_n$ be independent and identically distributed random variables with $U_i \sim \text{Uniform}(0, 1)$.

Define the random variable $Z = \max(U_1, U_2, \dots, U_n)$.

The cumulative distribution function (CDF) of Z is given by:

$$F_Z(x) = \begin{cases} 0 & x < 0 \\ ? & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

What is the correct expression to fill in the missing part of the CDF (i.e., for $0 \leq x \leq 1$)?

A. x^n

B. $1 - x^n$

C. nx^{n-1}

D. $1 - (1 - x)^n$

Your Answer:

Correct Answer: A

Not Attempted

Time taken: 00min 03sec

Discuss

Q #7**Multiple Choice Type****Award: 2****Penalty: 0.67****Probability**

Let X be a continuous random variable.

X has a constant PDF on $(0, 1]$, and another constant PDF on $(1, 2]$.

The PDF of X is given by:

$$f_X(x) = \begin{cases} c & \text{for } 0 < x \leq 1 \\ d & \text{for } 1 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

It is known that

$$\mathbb{P}(1 \leq X \leq 2) = 2 \cdot \mathbb{P}(0 \leq X < 1)$$

What are the values of c and d ?

- A. $c = 1/2, \quad d = 1/2$
- B. $c = 1/3, \quad d = 2/3$
- C. $c = 2/3, \quad d = 1/3$
- D. $c = 1/4, \quad d = 3/4$

Your Answer:**Correct Answer: B****Not Attempted****Time taken: 00min 04sec****Discuss**

Q #8

Multiple Choice Type

Award: 2

Penalty: 0.67

Probability

If X, Y , and Z are independent random variables taking values in $\mathbb{Z}$ (i.e., only integer values), what's the probability that $X + Y = 9$ and $Y - Z = 5$?

A. $\sum_{k \in \mathbb{Z}} p_X(9 + k) \cdot p_Y(k) \cdot p_Z(k + 5)$

B. $\sum_{k \in \mathbb{Z}} p_X(9 - k) \cdot p_Y(k) \cdot p_Z(k - 5)$

C. $\sum_{k \in \mathbb{Z}} p_X(9 - k) \cdot p_Y(k) \cdot p_Z(k + 5)$

D. $\sum_{k \in \mathbb{Z}} p_X(9 + k) \cdot p_Y(k) \cdot p_Z(k - 5)$

Your Answer:

Correct Answer: B

Not Attempted

Time taken: 00min 04sec

Discuss

Q #9

Multiple Choice Type

Award: 2

Penalty: 0.67

Probability

Suppose that X and Y are discrete random variables taking values in $\{1, 2\}$, with joint distribution given by:

$$\begin{aligned}\Pr[X = 1, Y = 1] &= \frac{1}{2}, & \Pr[X = 1, Y = 2] &= \frac{1}{10} \\ \Pr[X = 2, Y = 1] &= \frac{1}{10}, & \Pr[X = 2, Y = 2] &= \frac{3}{10}\end{aligned}$$

What is the value of $\mathbb{E}[X \mid Y = 1]$?

A. $3/2$

B. $6/5$

C. $7/6$

D. $5/4$

Your Answer:

Correct Answer: C

Not Attempted

Time taken: 00min 03sec

Discuss

Q #10

Multiple Select Type

Award: 2

Penalty: 0

Probability

The joint probability density function of X and Y is given by

$$f(x, y) = \begin{cases} c(x + y) & \text{if } 0 \leq x, y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

for some constant c . Which of the following statements is/are correct?

A. $c = 1$

B. X and Y are independent

C. The probability density function of X is $g(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$

D. $X + Y$ has a probability density function

Your Answer:

Correct Answer: A;D

Not Attempted

Time taken: 00min 02sec

Discuss

Q #11

Multiple Choice Type

Award: 2

Penalty: 0.67

Probability

Let $X_1, X_2, \dots, X_{2n}$ be random variables such that $V(X_i) = 4, i = 1, 2, \dots, 2n$ and $\text{Cov}(X_i, X_j) = 3, 1 \leq i \neq j \leq 2n$. Then

$V(X_1 - X_2 + X_3 - X_4 + \dots + X_{2n-1} - X_{2n})$ is :

A. n

B. $2n$

C. $3n - 2$

D. $n + 1$

Your Answer:

Correct Answer: B

Not Attempted

Time taken: 00min 01sec

Discuss

Q #12

Multiple Choice Type

Award: 2

Penalty: 0.67

Probability

Let X be Poisson(2) and Y be Binomial(10,3/4) random variables. If X and Y are independent, then $P(XY = 0)$ is

A. $e^{-2} + \left(\frac{1}{4}\right)^{10} (1 - e^{-2})$

B. $e^{-2} + \left(\frac{1}{4}\right)^{10} (1 - 2e^{-2})$

C. $e^{-2} \left(\frac{1}{4}\right)^{10}$

D. $e^{-2} + 1 - \left(\frac{1}{4}\right)^{10}$

Your Answer:

Correct Answer: A

Not Attempted

Time taken: 00min 01sec

Discuss

Q #13

Multiple Choice Type

Award: 2

Penalty: 0.67

Probability

Let X and Y denote the lifetimes (in years) of two independent components connected in a series with respective probability density functions

$$f_X(x) = \frac{1}{2}e^{-\frac{x}{2}}, x > 0 \quad \text{and} \quad f_Y(y) = \frac{y}{4}e^{-\frac{y}{2}}, y > 0$$

Then the probability that the system will survive for at least 2 years, is

A. e^{-2}

B. $2e^{-2}$

C. $3e^{-2}$

D. $4e^{-2}$.

Your Answer:

Correct Answer: B

Not Attempted

Time taken: 00min 00sec

Discuss

Q #14

Multiple Choice Type

Award: 2

Penalty: 0.67

Probability

Let Y be an exponential random variable with mean $1/\theta$, where $\theta > 0$. The conditional distribution of X given Y has Poisson distribution with mean Y . Then, the variance of X is

A. $\frac{1}{\theta^2}$

B. $\frac{\theta+1}{\theta}$

C. $\frac{\theta^2+1}{\theta^2}$

D. $\frac{\theta+1}{\theta^2}$

Your Answer:

Correct Answer: D

Not Attempted

Time taken: 00min 01sec

Discuss

Q #15

Numerical Type

Award: 2

Penalty: 0

Probability

Activities A , B , C and D form the critical path for a project with a PERT network. The means and variances of the activity duration for each activity are given below. All activity durations follow the Gaussian (normal) distribution, and are independent of each other.

Activity	A	B	C	D
Mean(days)	6	11	8	15
Variance(days ²)	4	9	4	9

The probability that the project will be completed within 40 days is :

(round off to two decimal places).

(Note: Probability is a number between 0 and 1).

Your Answer:

Correct Answer: 0.5

Not Attempted

Time taken: 00min 00sec

Discuss